

MA1521 L1 Week 05 Wed 20260909

Gradient (vector) of a function $f = (f_x \ f_y \ f_z)^t = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$
 "nabla f" ∇f transpose

In each of the following, a function f , a point P , and a vector A are given. Calculate the gradient of f at the point, and the directional derivative $\left. \frac{df}{ds} \right|_u$ at the point, in the direction u of the given vector A .

- a) $f(x,y) = x^3 + 2y^3$; $P = (1,1)$, $A = i - j$
 b) $w(x,y,z) = \frac{xy}{z}$; $P = (2,-1,1)$, $A = i + 2j - 2k$
 c) $z = x \sin y + y \cos x$; $P = (0, \pi/2)$, $A = -3i + 4j$
 d) $f(u,v,w) = (u + 2v + 3w)^2$; $P = (1,-1,1)$, $A = -2i + 2j - k$

a) $\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 3x^2 \\ 6y^2 \end{pmatrix}$ At point $P = (1,1)$, $\nabla f(1,1) = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

vector-valued function
of (x,y)

vector dot prod

Directional derivative $\left. \frac{df}{ds} \right|_u$ at point $P = (\nabla f)_P \cdot u$
 unit vector

$\left. \frac{df}{ds} \right|_u \text{ at } P = \nabla f(1,1) \cdot \frac{\vec{A}}{|\vec{A}|} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \cdot \frac{\begin{pmatrix} 1 \\ -1 \end{pmatrix}}{\sqrt{2}} = \frac{-3}{\sqrt{2}}$

b) $\nabla w = \begin{pmatrix} \frac{\partial w}{\partial x} \\ \frac{\partial w}{\partial y} \\ \frac{\partial w}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{y}{z} \\ \frac{x}{z} \\ -\frac{xy}{z^2} \end{pmatrix}$ At point $P = (2, -1, 1)$, $\nabla f(2, -1, 1) = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$

$$\left. \frac{dw}{ds} \right|_{\mathbf{u}} \text{ at } P = \nabla f(2, -1, 1) \cdot \frac{\vec{A}}{|\vec{A}|} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \cdot \frac{\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}}{\sqrt{9}} = \frac{-1}{3}$$

c) $\nabla z = \begin{pmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{pmatrix} = \begin{pmatrix} \sin y - y \sin x \\ x \cos y + \cos x \end{pmatrix}$ At point $P = (0, \frac{\pi}{2})$, $\nabla z(0, \frac{\pi}{2}) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$\left. \frac{dz}{ds} \right|_{\mathbf{u}} \text{ at } P = \nabla z(0, \frac{\pi}{2}) \cdot \frac{\vec{A}}{|\vec{A}|} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{\begin{pmatrix} -3 \\ 4 \end{pmatrix}}{\sqrt{25}} = \frac{1}{5}$$

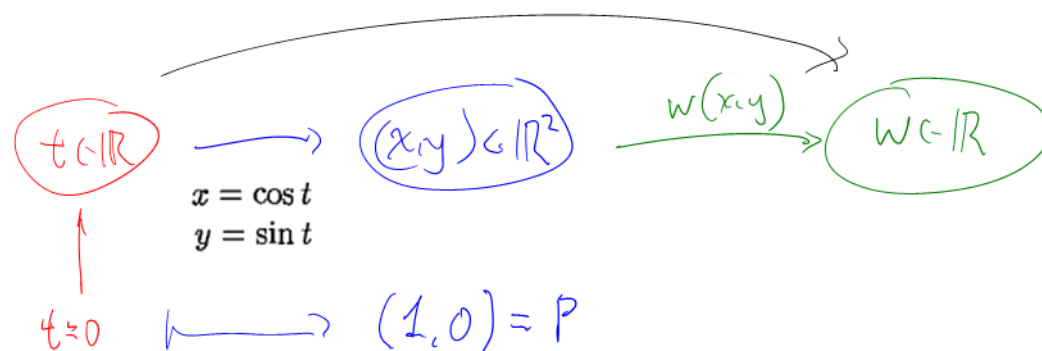
d) $\nabla f = \begin{pmatrix} f_u \\ f_v \\ f_w \end{pmatrix} = \begin{pmatrix} 2(u+2v+3w) \cdot 1 \\ 2(u+2v+3w) \cdot 2 \\ 2(u+2v+3w) \cdot 3 \end{pmatrix}$ At point $P = (1, -1, 1)$, $\nabla f(1, -1, 1) = \begin{pmatrix} 4 \cdot 1 \\ 4 \cdot 2 \\ 4 \cdot 3 \end{pmatrix} = 4 \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$$\left. \frac{df}{ds} \right|_{\mathbf{u}} \text{ at } P = \nabla f(1, -1, 1) \cdot \frac{\vec{A}}{|\vec{A}|} = 4 \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \frac{\begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}}{\sqrt{9}} = \frac{-4}{3}$$

In the following, $x = \cos t$, $y = \sin t$, and $w = f(x(t), y(t))$ where $f(x, y)$ is an unknown function

a) Suppose $\nabla w = 2\mathbf{i} + 3\mathbf{j}$ at $P : (1, 0)$; Find the value of $\frac{dw}{dt}$ at $\underline{t = 0}$.

b) Suppose $\nabla w = y\mathbf{i} + x\mathbf{j}$; Find $\frac{dw}{dt}$ and tell for what t -values it is zero.



a) know $(\nabla w)|_{P=(1,0)} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. want $\frac{dw}{dt}\bigg|_{t=0} = ?$

Chain rule, $\frac{dw}{dt} = \begin{pmatrix} w_x & w_y \end{pmatrix} \cdot \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}$

At $P=(1,0)$, $\frac{dw}{dt}\bigg|_{t=0} = (\nabla w)|_{P=(1,0)} \cdot \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}\bigg|_{t=0} = \begin{pmatrix} 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 3$

$\downarrow$
 $t=0$

b) Know $\nabla w = \begin{pmatrix} y \\ x \end{pmatrix} \iff (w_x \ w_y) = (y \ x)$

Want $\frac{dw}{dt}$? When is it 0 ?

Chain rule, $\frac{dw}{dt} = (w_x \ w_y) \cdot \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} = (y \ x) \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$
 $= -y \sin t + x \cos t$

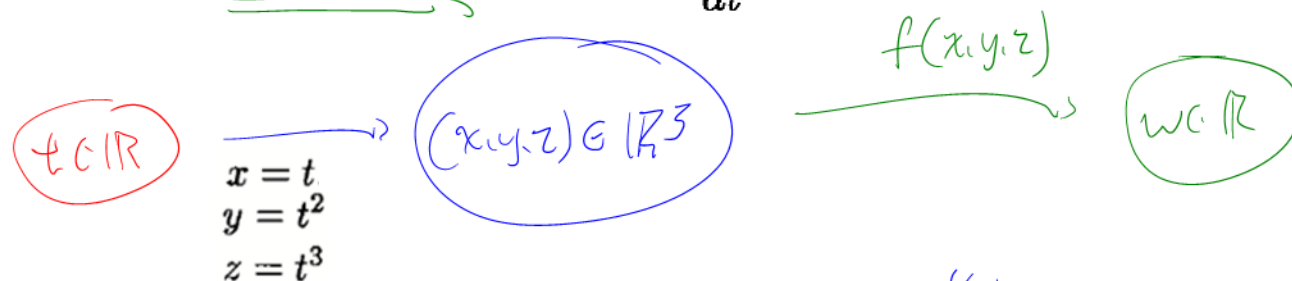
using $\begin{matrix} x = \cos t \\ y = \sin t \end{matrix} = -\sin^2 t + \cos^2 t = \underline{\underline{\cos 2t}}$

Hence $\frac{dw}{dt} = 0$ iff $\cos 2t = 0$ iff $t = \frac{\pi}{4} + \frac{n\pi}{2}$, $n \in \mathbb{Z}$

In the following, $x = t$, $y = t^2$, $z = t^3$, and $f(x, y, z)$ is an unknown function

a) Suppose $\nabla f = \langle \underline{1, -1, 2} \rangle$ at $\underline{(1, 1, 1)}$. Find $\frac{df}{dt}$ at $\underline{t = 1}$.

b) Suppose $\nabla f = \langle \underline{3x^2y, x^3 + z, y} \rangle$; Find $\frac{df}{dt}$



Chain rule:
$$\frac{df}{dt} = \begin{pmatrix} f_x & f_y & f_z \end{pmatrix} \cdot \begin{pmatrix} x'(t) \\ y'(t) \\ z'(t) \end{pmatrix} = (\nabla f) \cdot \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix}$$

a) $t = 1 \rightarrow (1, 1, 1)$

Hence
$$\left. \frac{df}{dt} \right|_{t=1} = \left. (\nabla f) \right|_{(1,1,1)} \cdot \left. \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix} \right|_{t=1} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 5$$

b)
$$\frac{df}{dt} = (\nabla f) \cdot \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix} = \begin{pmatrix} 3x^2y \\ x^3 + z \\ y \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix} = \begin{pmatrix} 3t^4 \\ 2t^3 \\ t^2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2t \\ 3t^2 \end{pmatrix} = 10t^4$$

For the following functions, each with a given point P ,

- (i) find the maximum and minimum values of $\left. \frac{dw}{ds} \right|_{\mathbf{u}}$, as $\mathbf{u}$ varies;
- (ii) tell for which directions the maximum and minimum occur;
- (iii) find the direction(s) $\mathbf{u}$ for which $\left. \frac{dw}{ds} \right|_{\mathbf{u}} = 0$.

a) $w = \ln(3x - 4y), \quad (1, 1)$

b) $w = xy + yz + xz, \quad (1, -1, 2)$

a) $\nabla w = \begin{pmatrix} w_x \\ w_y \end{pmatrix} = \frac{1}{3x-4y} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ At point $P=(1,1)$, $(\nabla w)_P = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$

$\left(\left. \frac{dw}{ds} \right|_{\mathbf{u}} \right)_P = \underbrace{(\nabla w)_P \cdot \mathbf{u}}_{|\nabla w)_P| |\mathbf{u}| \cos(\text{angle betw})}$ is $\begin{cases} \text{max} \\ \text{min} \\ 0 \end{cases}$ when angle betw $(\nabla w)_P$ & $\mathbf{u}$ is $\begin{cases} 0 \\ \pi \\ \pm \pi/2 \end{cases}$

$\begin{cases} \text{max} \\ \text{min} \end{cases}$ of value $\begin{cases} 5 \\ -5 \end{cases}$ in direction $\begin{cases} -\frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix} \\ \frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix} \end{cases}$ i.e. when $\mathbf{u}$ is $\begin{cases} (\nabla w)_P / |(\nabla w)_P| = \frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix} \\ -(\nabla w)_P / |(\nabla w)_P| = -\frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix} \\ \pm \frac{1}{5} \begin{pmatrix} 4 \\ 3 \end{pmatrix} \end{cases}$

(iii) $\left. \frac{dw}{ds} \right|_{\mathbf{u}} = 0$ in direction $\pm \frac{1}{5} \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

b) $\nabla w = \begin{pmatrix} w_x \\ w_y \\ w_z \end{pmatrix} = \begin{pmatrix} y+z \\ x+z \\ y+x \end{pmatrix}$ At point $P = (1, -1, 2)$, $(\nabla w)_P = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$

$\left(\frac{dw}{ds}\right)_P = \underbrace{(\nabla w)_P \cdot \mathbf{u}}_{|(\nabla w)_P| |\mathbf{u}| \cos(\text{angle betw})}$ is $\begin{cases} \text{max} \\ \text{min} \\ 0 \end{cases}$ when angle betw $(\nabla w)_P$ & $\mathbf{u}$ is $\begin{cases} 0 \\ \pi \\ \pm \pi/2 \end{cases}$

i.e. when $\mathbf{u}$ is $\begin{cases} (\nabla w)_P / |(\nabla w)_P| = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \\ -(\nabla w)_P / |(\nabla w)_P| = -\frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \end{cases}$

$\begin{cases} \text{max} \\ \text{min} \end{cases}$ value of $\left(\frac{dw}{ds}\right)_P$ is $\begin{cases} \sqrt{10} \\ -\sqrt{10} \end{cases}$ in direction $\begin{cases} \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \\ -\frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \end{cases}$

(iii) Want $\mathbf{u}$ to be $\perp$ to $(\nabla w)_P = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$
 $\parallel$ unit vector

$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ i.e. want $1 \cdot a + 3 \cdot b + 0 \cdot c = 0$ & $\sqrt{a^2 + b^2 + c^2} = 1$

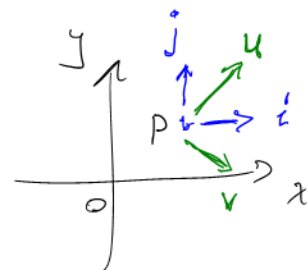
$\mathbf{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ such that $a + 3b = 0$ & $10b^2 + c^2 = 1$
 $(a = -3b)$

One example is $\mathbf{u} = \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} / \sqrt{10}$

Suppose $\frac{dw}{ds}\Big|_{\mathbf{u}} = 2$, $\frac{dw}{ds}\Big|_{\mathbf{v}} = 1$ at P , where $\mathbf{u} = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{2}}$, $\mathbf{v} = \frac{\mathbf{i} - \mathbf{j}}{\sqrt{2}}$. Find $(\nabla w)_P$.

$$w = f(x, y)$$

$$\nabla w = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad \text{At } P, \quad (\nabla w)_P = \begin{pmatrix} (f_x)_P \\ (f_y)_P \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$



$$\frac{dw}{ds}\Big|_{\mathbf{u}} = 2 \quad \text{at } P$$

means

$$(\nabla w)_P \cdot \mathbf{u} = 2$$

i.e.,

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2$$

$$\frac{dw}{ds}\Big|_{\mathbf{v}} = 1 \quad \text{at } P$$

$$(\nabla w)_P \cdot \mathbf{v} = 1$$

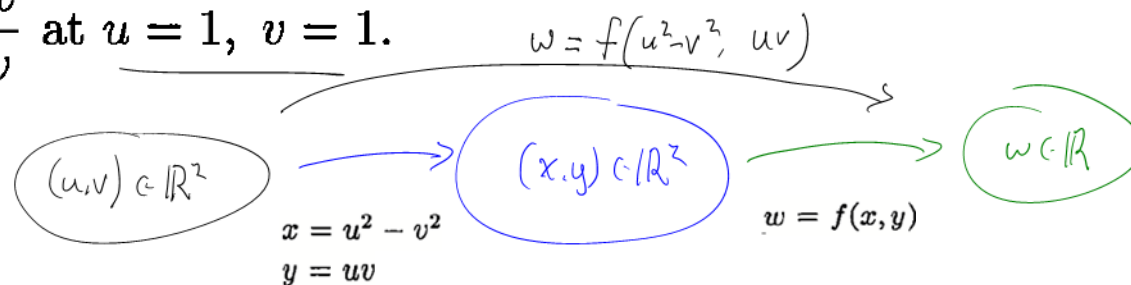
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1$$

$$\text{i.e.,} \quad \begin{aligned} a + b &= 2\sqrt{2} \\ a - b &= \sqrt{2} \end{aligned}$$

$$\text{Solve to get} \quad \begin{aligned} a &= \frac{3}{2}\sqrt{2} \\ b &= \frac{\sqrt{2}}{2} \end{aligned}$$

$$\text{So } (\nabla w)_P = \begin{pmatrix} \frac{3}{2}\sqrt{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Let $w = f(x, y)$, and assume that $\nabla w = 2\mathbf{i} + 3\mathbf{j}$ at $(0, 1)$. If $x = u^2 - v^2$, $y = uv$, find $\frac{\partial w}{\partial u}$, $\frac{\partial w}{\partial v}$ at $u = 1$, $v = 1$.



$$\begin{matrix} u=1 \\ v=1 \end{matrix} \quad (1, 1) \quad \longrightarrow \quad (0, 1) \quad \begin{matrix} x=0 \\ y=1 \end{matrix}$$

Chain rule

$$\begin{pmatrix} \frac{\partial w}{\partial u} & \frac{\partial w}{\partial v} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}}_{(\nabla w)^t} \cdot \underbrace{\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}}_{\begin{pmatrix} 2u & -2v \\ v & u \end{pmatrix}}$$

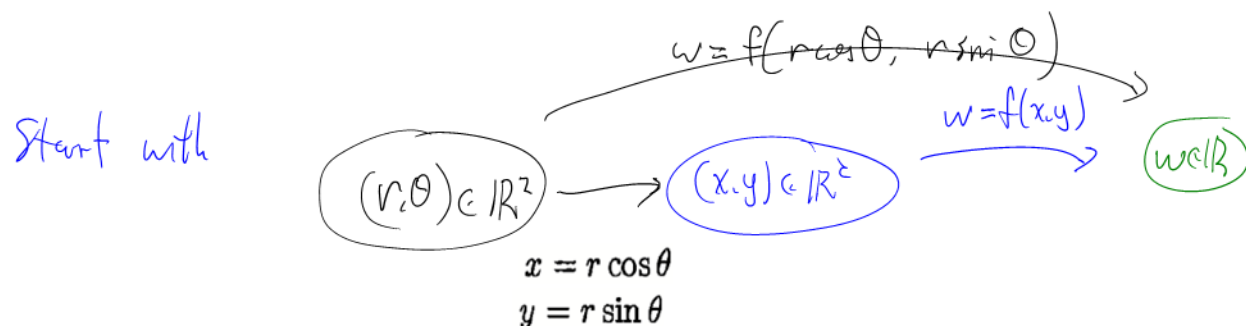
$$\begin{aligned} \text{At } \begin{matrix} u=1 \\ v=1 \end{matrix} \quad \begin{pmatrix} \frac{\partial w}{\partial u} & \frac{\partial w}{\partial v} \end{pmatrix}_{(1,1)} &= \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}_{\substack{(0,1) \\ x' \ y'}} \cdot \begin{pmatrix} 2u & -2v \\ v & u \end{pmatrix}_{\substack{(1,1) \\ u' \ v'}} \\ &= (2 \ 3) \cdot \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} = (7 \ -1) \end{aligned}$$

Let $w = f(x, y)$, and suppose we change from rectangular to polar coordinates:

$$\underline{x = r \cos \theta}, \quad \underline{y = r \sin \theta}.$$

- a) Show that $(w_x)^2 + (w_y)^2 = (w_r)^2 + \frac{1}{r^2} (w_\theta)^2$.
- b) Suppose $\nabla w = 2\mathbf{i} - \mathbf{j}$ at the point $x = 1, y = 1$.

Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial \theta}$ when $r = \sqrt{2}, \theta = \pi/4$



a) Chain rule gives $\begin{pmatrix} \frac{\partial w}{\partial r} & \frac{\partial w}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}$

i.e. $\begin{pmatrix} w_r & w_\theta \end{pmatrix} = \begin{pmatrix} w_x & w_y \end{pmatrix} \cdot \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$ — ~~(*)~~

$$= \begin{pmatrix} w_x \cos \theta + w_y \sin \theta & -w_x r \sin \theta + w_y r \cos \theta \end{pmatrix}$$

Thus

$$\begin{aligned}
 (w_r)^2 + \frac{1}{r^2} (w_\theta)^2 &= \left(w_x \cos \theta + w_y \sin \theta \right)^2 + \frac{1}{r^2} \left(-w_x r \sin \theta + w_y r \cos \theta \right)^2 \\
 &= w_x^2 \cos^2 \theta + w_y^2 \sin^2 \theta + w_x^2 \sin^2 \theta + w_y^2 \cos^2 \theta \\
 &\quad + 2 w_x w_y \sin \theta \cos \theta - 2 w_x w_y \sin \theta \cos \theta \\
 &= (w_x^2 + w_y^2) (\cos^2 \theta + \sin^2 \theta) = (w_x)^2 + (w_y)^2
 \end{aligned}$$

b) First note that the point

$P: \begin{cases} r = \sqrt{2} \\ \theta = \pi/4 \end{cases}$ is mapped via $\begin{matrix} x = r \cos \theta \\ y = r \sin \theta \end{matrix}$ to the point $Q: \begin{cases} x = 1 \\ y = 1 \end{cases}$

We are told that $(\nabla w) = 2\mathbf{i} - \mathbf{j}$ at the point P

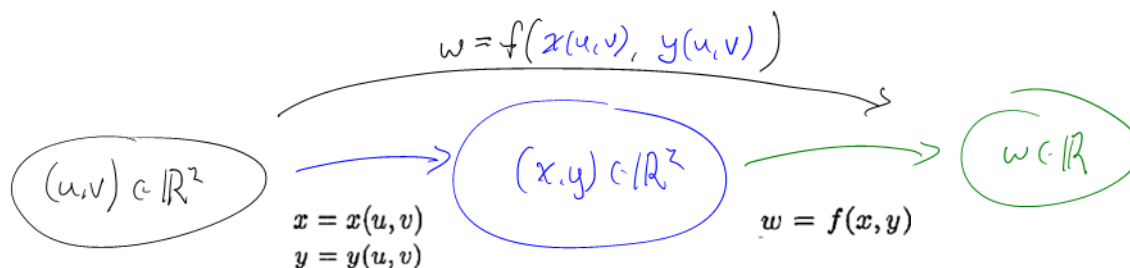
i.e. $(w_x \ w_y)|_Q = \begin{pmatrix} 2 & -1 \end{pmatrix}$

Equation (*) at point P now gives

$$\begin{aligned}
 (w_r \ w_\theta)|_P &= (w_x \ w_y)|_Q \cdot \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} \Big|_P \\
 &= \begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} \cos \frac{\pi}{4} & -\sqrt{2} \sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \sqrt{2} \cos \frac{\pi}{4} \end{pmatrix} \\
 &= \begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & -1 \\ \frac{1}{\sqrt{2}} & 1 \end{pmatrix} = \underline{\underline{\begin{pmatrix} \frac{1}{\sqrt{2}} & -3 \end{pmatrix}}}
 \end{aligned}$$

The Jacobian matrix for the change of variables $x = x(u, v)$, $y = y(u, v)$ is defined to be $J = \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$. Let $\nabla f(x, y)$ be represented as the row vector $\langle f_x \ f_y \rangle$. Show that

$$\nabla f(x(u, v), y(u, v)) = \nabla f(x, y) \cdot J \quad (\text{matrix multiplication}).$$



Chain rule :

$$\begin{pmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

$$= \langle f_x \ f_y \rangle \cdot \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$$

i.e., $\nabla f(x(u, v), y(u, v)) = \nabla f(x, y) \cdot J$